MINING DEALER BEST PRACTICE

Eliminating Live Work When Purging Air from the Brake System on 785C Trucks

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

June 2024

1.11-240621-1

1.0 Introduction

Live Work is any task (i.e testing, commissioning, or troubleshooting) that requires a system to be energized and the person performing the job to enter the footprint of the machine **(excludes doing the same task from the machine cabin or outside the machine footprint).**

Caterpillar has worked closely with the mining industry to review service literature and maintenance and repair procedures as well as testing and adjustment tasks to find solutions to remove live work from these publications. Dealers are also developing procedures to complete maintenance in more clever and safer ways and this best practice is a good example of that.

Brake purging on LMTs is a key part of the maintenance activities on a mine site and is done regularly as per the operating and maintenance manual. This procedure is described in SENR1494.This document does not specify the tools to produce quick couplers on each 4 corners to facilitate the air bleeding and the hose assembly needs to perform this activity outside the footprint of the machine while maintaining eye contact for the operator.

2.0 Best Practice Description

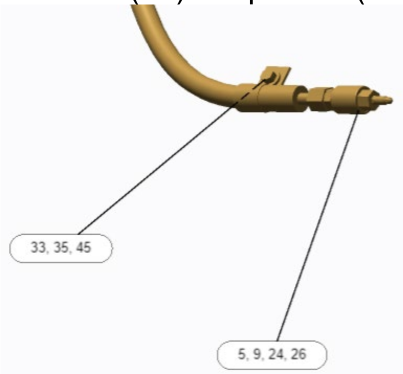
This best practice covers the fabrication of a ‘’brake air purge tool GP’’ as well as modifications to the rear final axle and front wheel hubs to facilitate the maintenance task for bleeding the air from the brake system.

3.0 Implementation Steps

This best practice complements SENR1494.
All steps in this instruction apply. The difference is that the drain valves have been made more accessible, with a quick-connect to facilitate this task and that the drain hose is fabricated with appropriate length to allow eye contact with the operator.

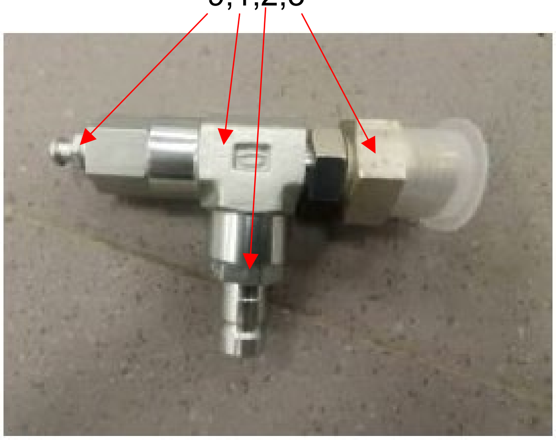
Rear Axle Installation

1. Remove (26) adapter as (169-7403) and (9) Purge screw (9F-2167) from hose as (152-7744).



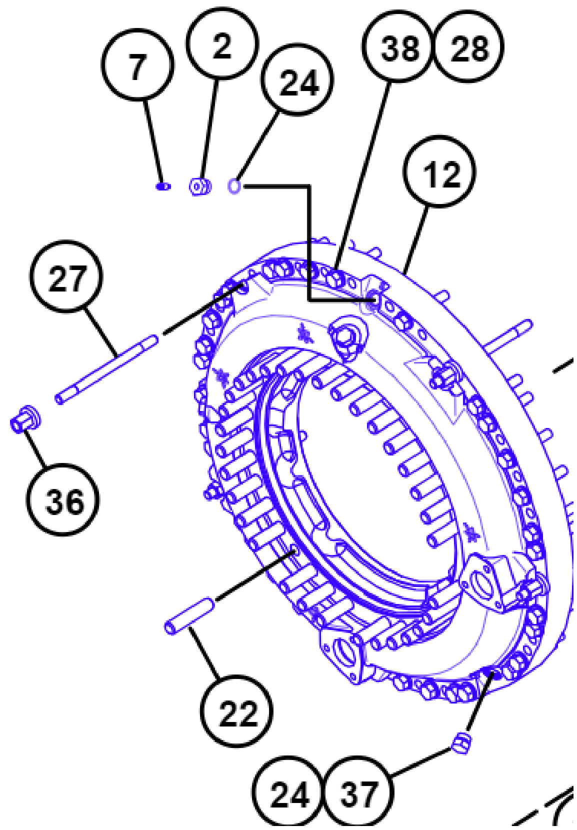
2. Install (1) Tee (3E-7410), (3) Adapter (6G-3432), (2) Quick Connect Tap (6V-3965), Seal (2M-9780), Seal (3J-1907), Seal (214-7568), (9) Purge Screw (9F-2167), Cap- Dust (6V-0852).

9,1,2,3

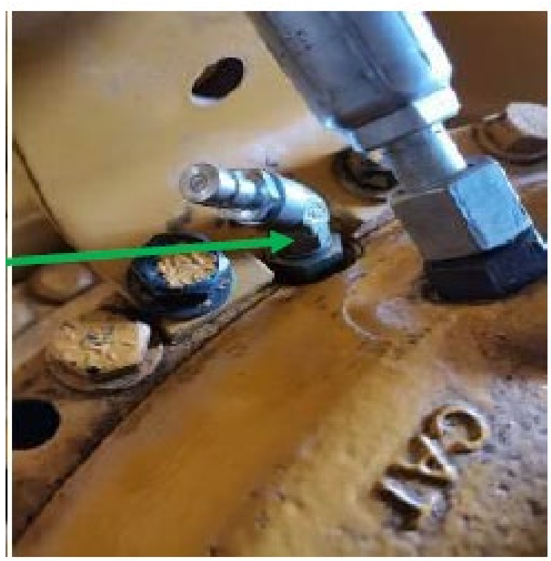


Front Hub Installation

1. Remove (2) Adapter (9D-6125) and (7) Purge Screw (9F-2167) from both front brake Anchors (177-1606).



2. Install (F1) Adapter (5P-1404), (F2) Elbow (6V-8077), (F3) Quick Connect Tap (6V-3965), (F4) Seal (214-7568), (F5) Cap- Dust (6V-0852) to both front brake anchors (177-1606).



Air purge tool GP Fabrication.

1. Build air Purge tool using below parts.

Required Tools/Parts			
Item	Qty	Part Number	Part Name
T1	1	9F-2167	Purge Screw
T2	1	6G-3432	Adapter
T3	1	3J-1907	Seal-O-Ring
T4	1	4J-5477	Seal-O-Ring
T5	1	6V-8398	Seal-O-Ring
T6	1	8W-0222	Adapter
T7	1	6V-9873	Adapter
Hose1			
H1	1	124-1891	Coupling
H2	1	154-0095	Coupling
H3	8m	4568426	Hose Bulk

Brake Air Purge

1. Use a **177-7862** Hose As and two **6V-4143** Coupler Assemblies to construct a bypass hose.
2. Apply the parking brake. This will release the pressure in the lines.

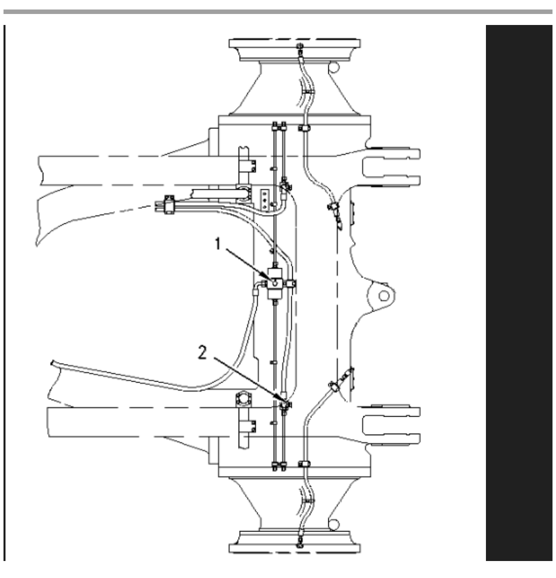


Illustration 1 g00297437
(1) Pressure tap on the rear slack adjuster
(2) Pressure tap for parking brake release pressure

3. Put one end of the bypass hose on the pressure tap for the slack adjuster. Use the pressure tap on the rear slack adjuster (1) for both rear wheels. Use the pressure tap on the front slack adjuster for both front wheels.
4. Put the other end of the bypass hose on a pressure tap for parking brake release pressure (2).

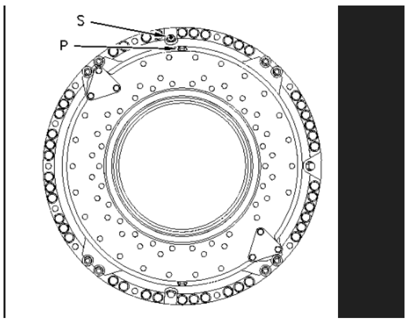


Illustration 2 g00282133
(P) Purge valve for the parking brake
(S) Purge valve for the service brake

5. **Note:** Identification letters are embossed on the purge valves at the top of the brake housing. The purge valve for the parking brake is labelled with a "P". The purge valve for the service brake is labelled with an "S".
6. Connect air purge tool to the Quick Connect Tap (6V-3965) on either of the rear wheels to a suitable container.
7. Run the engine at low idle. Move the parking brake lever to the OFF position.
8. Pull down the retarder control.
9. Open the purge valve (9F-2167) on the Air Purge Tool GP. Leave the purge valve open until there is a constant flow of oil.
10. Close the purge valve. Move the retarder control to the OFF position. Activate the parking brake to release the pressure. Remove the jumper hose.

    Note: To verify that all the air was removed from the service brake lines, perform the following steps.

    Note: To make sure that the makeup oil tank is full, operate the engine at high idle for 20 seconds before each brake application.

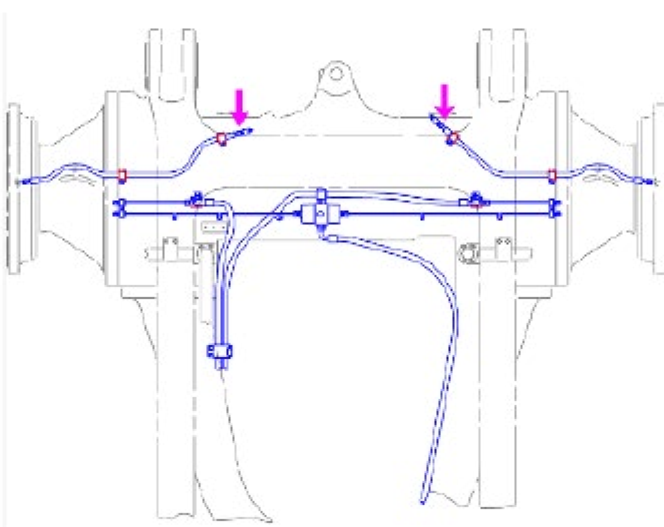
11. Move the parking brake lever to the OFF position to allow full travel of the service brake piston.
12. Push down on the brake pedal or pull down the retarder control.
13. Open the purge valve (9F-2167) on the Air Purge Tool on a front wheel brake housing and purge the brake system. Leave the purge valve open until there is a continuous flow of oil.
14. Close the purge valve. Release the brake pedal or move the retarder control to the OFF position.
15. Do Steps 11 through 13 until there is a continuous flow of oil.
16. Do Steps 1 through 14 for the other three wheels.

Note: After you purge the brakes, you may need to reset the actuating pins for the overstroke switch. The actuating pins for the overstroke switch are on the master cylinders. A brake overstroke condition is a Level III warning.

When you need to reset the actuating pin, you will push the actuating pin into the housing. Push the actuating pin into the housing until the bottom of the pin is flush with the bottom of the housing.

4.0 Benefits

When completing the procedure, the air purge tool GP can be connected to the quick connect modifications for each wheel to perform this task quicker and safer by having direct visual contact with the operator in the cab. Below example shows the purge of air from one of the rear wheels drain points:



5.0 Resources Required

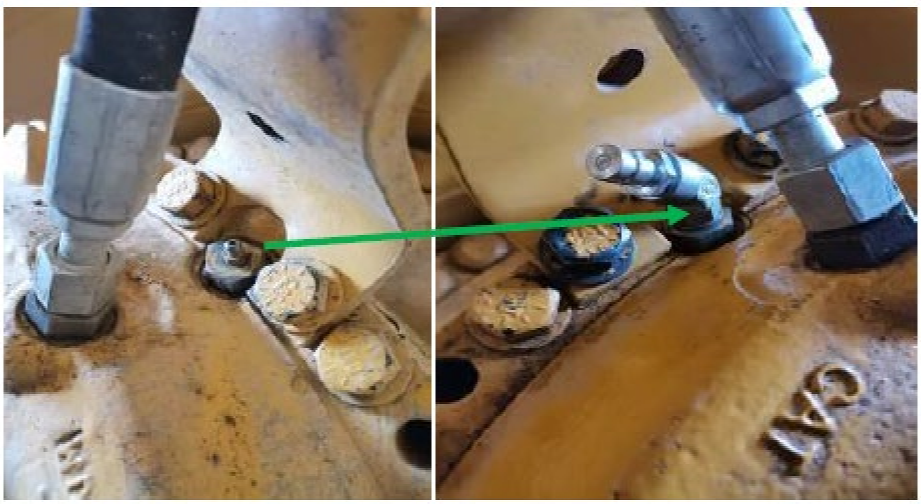
5.1 People

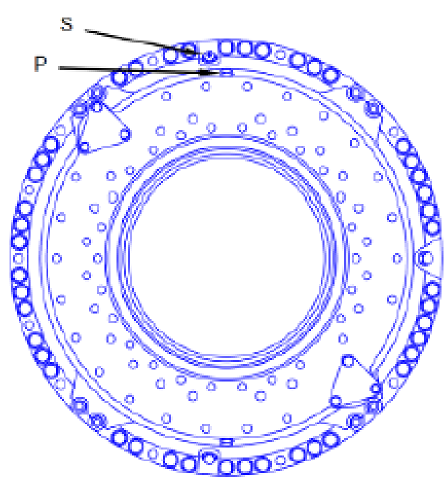
Job can be performed by two people (Operator and technician).

5.2 Tooling

FRONT HUBS

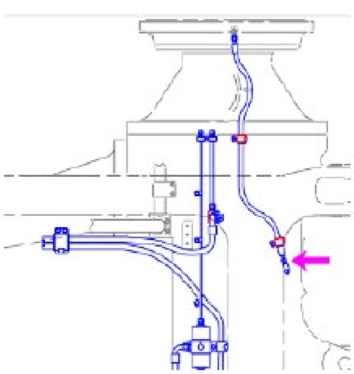
Required Tools			
Item	Qty	Part Number	Part Name
Front			
F1	2	5P-1404	Adapter
F2	2	6V-8077	Elbow
F3	2	6V-3965	Quick Connect Tap
F4	2	214-7568	Seal-O-Ring
F5	2	6V-0852	Cap-Dust

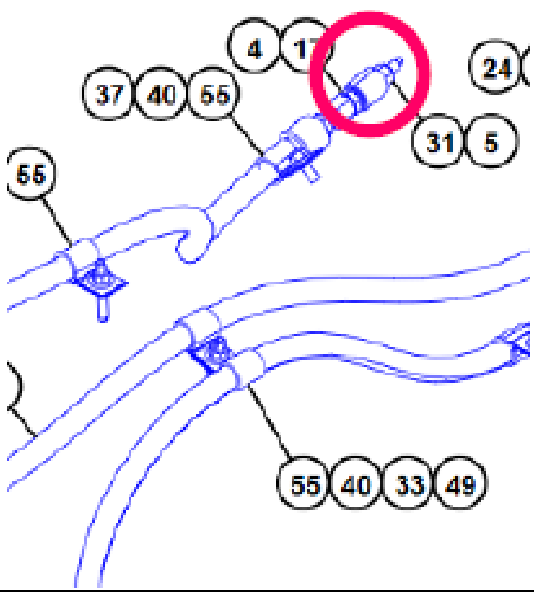




REAR WHEELS

Required Tools			
Item	Qty	Part Number	Part Name
Rear			
R1	2	3E-7410	Tee
R2	2	6G-3432	Adapter
R3	2	6V-3965	Quick Connect Tap
R4	2	2M-9780	Seal-O-Ring
R5	6	3J-1907	Seal-O-Ring
R6	2	214-7568	Seal-O-Ring
R7	2	9F-2167	Purge Screw
R8	2	6V-0852	Cap-Dust





''Air purge tool GP''

Required Tools/Parts			
Item	Qty	Part Number	Part Name
T1	1	9F-2167	Purge Screw
T2	1	6G-3432	Adapter
T3	1	3J-1907	Seal-O-Ring
T4	1	4J-5477	Seal-O-Ring
T5	1	6V-8398	Seal-O-Ring
T6	1	8W-0222	Adapter
T7	1	6V-9873	Adapter
Hose1			
H1	1	124-1891	Coupling
H2	1	154-0095	Coupling
H3	8m	4568426	Hose Bulk


By-pass hose (same as SENR1494)

Tools Needed		
Part Number	**Part Description**	**Qty**
177-7862	Hose As	1
6V-4143	Coupler	2

6.0 Supporting Attachments / References

SENR1494

7.0 Related Best Practices

Mining Dealer Best Practice
Maintenance & Repair: Elimination of Live Work - Checking HMS Slew Ring Tilting Play
(Publication Number 1.11-230525-1)

8.0 Acknowledgements

This procedure and tooling was developed by the Mantrac site team at the Sukari Gold Mine in Egypt.

Following a request from the customer to consider the elimination of live work when carrying out brake bleeding.

This best practice was written in collaboration between Angel Rincon Martinez CAT Industry Rep and Edward Nicklin, Mantrac Mining Support Performance Manager.